

# Classification Report

Created at: 2026-06-04T09:58

Tag: mineral\_wool\_01\_synth

## Dataset:

### Dataset summary:

| Split      | Count | Share of overall |
|------------|-------|------------------|
| train      | 3807  | 97.97%           |
| validation | 79    | 2.03%            |
| TOTAL      | 3886  | 100.00%          |

| Class      | Count (train) | Share of train | Count (validation) | Share of validation |
|------------|---------------|----------------|--------------------|---------------------|
| glass_wool | 2056          | 8.73%          | 46                 | 58.23%              |
| stone_wool | 1751          | 7.44%          | 33                 | 41.77%              |
| TOTAL      | 3807          | 16.17%         | 79                 | 100.00%             |

### Real vs. synthetic training split:

| Class      | Count (train, real) | Share of train (real) | Count (train, synthetic) | Share of train (synthetic) |
|------------|---------------------|-----------------------|--------------------------|----------------------------|
| glass_wool | 2056                | 100.00%               | 0                        | 0.00%                      |
| stone_wool | 1751                | 100.00%               | 0                        | 0.00%                      |
| total      | 3807                | 100.00%               | 0                        | 0.00%                      |

### Data file path:

[Data file](#)

# Model:

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## Model architecture:

| Component               | Description |
|-------------------------|-------------|
| Backbone                | resnet18    |
| Number of frozen layers | 7           |
| Pretrained              | True        |

## Training configuration:

| Hyperparameter        | Value     |
|-----------------------|-----------|
| Batch size            | 16        |
| Learning rate         | 0.0005    |
| Weight decay          | 0.01      |
| Epochs                | 100       |
| Device                | cuda      |
| Number of CPU workers | 8         |
| Image size            | 1920x1200 |

## Training plots:

### Training/ validation loss:

 Training and Validation Loss

### Training/ validation accuracy:

 Training and Validation Accuracy

## Model path:

[Model path](#)

# Evaluation:

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## Confusion matrix:

 Confusion Matrix (Simple)

## Confusion Matrix (Fancy)

Confidence thresholds:

### Classification report:

#### Overall metrics:

| Overall Metrics | Value |
|-----------------|-------|
| Accuracy        | 0.911 |
| Precision       | 0.947 |
| Recall          | 0.911 |
| F1 Score        | 0.929 |

#### Per-Class Metrics

| Class        | Precision | Recall | F1-Score | Support |
|--------------|-----------|--------|----------|---------|
| glass_wool   | 0.96      | 0.93   | 0.95     | 46      |
| stone_wool   | 0.94      | 0.88   | 0.91     | 33      |
| Unsure       | 0.00      | 0.00   | 0.00     | 0       |
|              |           |        |          |         |
| accuracy     |           |        | 0.91     |         |
| macro avg    | 0.63      | 0.60   | 0.62     | 79      |
| weighted avg | 0.95      | 0.91   | 0.93     | 79      |

#### Misclassifications:

#### Source files:

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#### Source files:

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- [Configuration file](#)
- [Data file](#)
- [Model file](#)
- [Inference Record CSV](#)
- [Evaluation metrics and plots](#)